Fiscal Year:	FY 2009	**Task Last Updated:**	FY 07/27/2009
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Analysis of a Novel Sensory Mechanism in Root Phototropism		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Greensboro	**State:**	NC
Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-02
Start Date:	10/01/2000	**End Date:**	07/31/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	ISS NOTE: Received extension to 07/31/2009 per A. Chu (6/08) NOTE: Received extension to 07/31/2008 per A. Chu (5/07) Note anniversary date changed to 8/1; start date remains the same (per A. Chu 4-06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Edelmann, Richard (Miami University) Correll, Melanie (University of Florida)		
Grant/Contract No.:	NCC2-1200		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The major goals of this research are (1) to better understand cellular mechanisms of phototropism in plants and (2) to determine the effects and influence of gravity on light perception in plants. Specifically, we plan to use microgravity to facilitate studies of phototropism in roots and hypocotyls of young seedlings of Arabidopsis. Results of our ground-based studies suggest that there are two photosensory systems that elicit phototropic responses in roots: a previously identified blue-light photoreceptor system and a red light phytochrome-based mechanism that we have recently discovered. The phototropic response in roots is much weaker than the graviresponse, which competes with and often masks the phototropic response. It was through the use of mutant plants with a weakened gravitropism that we were able to identify the activity of the red-light-dependent phototropic system. Because of the interfering effect of the strong gravitropic response in roots, microgravity conditions are needed to effectively study this novel red-light-dependent phototropic response. In this project, we utlized the European Modular Cultivation System (EMCS) on the ISS. The experiment was performed on the ISS Oct.-Dec. 2006. The last samples were returned to earth on STS-120 in Nov. 2007. This information is important for growing plants on long-term space missions as part of life support systems and advances the current exploration agenda at NASA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	From a basic science perspective, we will gain new insight into cellular/molecular events in light sensing. In the long term, through greater understanding of how plants perceive and respond to light, it will be possible to improve crop productivity. It also should be possible to modify agricultural crop plants with genetically improved phytochrome or their downstream signaling molecules to develop transgenic crop species with more fine tuned light-responsive developmental traits to better fit their ambient light environments.
Task Progress:	Hypothesis 1-. Phototropism is enhanced in microgravity. While seed germination was lower compared to the ground controls and to seeds stored for shorter periods in hardware, the seedlings that did germinate exhibited vigorous growth and appeared healthy in downlinks as well as in images captured on the video tapes. While analysis is in progress, sample data of the response to light in hypocotyls showed that there is phototropic curvature in response to unilateral red light. Interestingly, the response was of a greater magnitude in microgravity compared to the 1-g control ($P < 0.05$; Least Squares Method). This extends previous reports that found that positive phototropism in response to unilateral blue light was enhanced in wheat coleoptiles in microgravity conditions but only at some fluence rates. In addition, growth of seedlings was similar in microgravity compared to the 1-g control. Hypothesis 2-. Microgravity conditions will change global gene expression patterns. Our preliminary results from the gene profiling experiments with the frozen plant samples indicate that genes in the major plant hormone pathways are affected by microgravity: auxin, cytokinin, ethylene, & brassionsteriods. In addition, genes in important developmental pathways are affected by microgravity such as photo-perception, heat shock, plastid development, root hairs, among others. In summary, while the TROPI experiment had technical and logistical difficulties, most of the procedures worked well, resulting in healthy plant seedlings that exhibited robust tropistic curvature. Procedures also were continually refined throughout the course of the project. Data analyses are in progress, but the initial results suggest that TROPI will yield new insights regarding plant development in microgravity, the relationship between gravitropism and phototropism, and global gene expression in altered gravity and light environments.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
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